US 20250283729A1

(19) United States
(12) Patent Application Publication
(10) Pub. No.: US 2025/0283729 A1
(43) Pub. Date: Sep. 11, 2025

KAWASE et al.

(54) **INFORMATION PROCESSING DEVICE, CONTENT PROVIDING METHOD, VEHICLE, AND PROGRAM**

(71) Applicant: **TOYOTA BOSHOKU KABUSHIKI KAISHA**, Aichi (JP)

(72) Inventors: **Osamu KAWASE**, Aichi (JP); **Nobuki HAYASHI**, Aichi (JP); **Yuki KAMIYA**, Aichi (JP); **Hiroki ASHIZAWA**, Aichi (JP); **Tetsuro SAKAGUCHI**, Aichi (JP); **Tomonori ISHIKAWA**, Aichi (JP)

(73) Assignee: **TOYOTA BOSHOKU KABUSHIKI KAISHA**, Aichi (JP)

(21) Appl. No.: **19/016,526**

(22) Filed: **Jan. 10, 2025**

(30) **Foreign Application Priority Data**

Mar. 8, 2024 (JP) 2024-036167

Publication Classification

(51) **Int. Cl.**
G01C 21/36 (2006.01)

(52) **U.S. Cl.**
CPC *G01C 21/3655* (2013.01); *G01C 21/3644* (2013.01)

(57) **ABSTRACT**

An information processing device includes: a setting section configured to set, in a route, a provision start point at which provision of content is started; a factor information obtaining section configured to obtain, while a traveling body is traveling along the route, factor information about a factor which gives an influence on time when the traveling body arrives at the provision start point; and an adjusting section configured to adjust, while the traveling body is traveling along the route, the provision start point on the basis of the factor information.

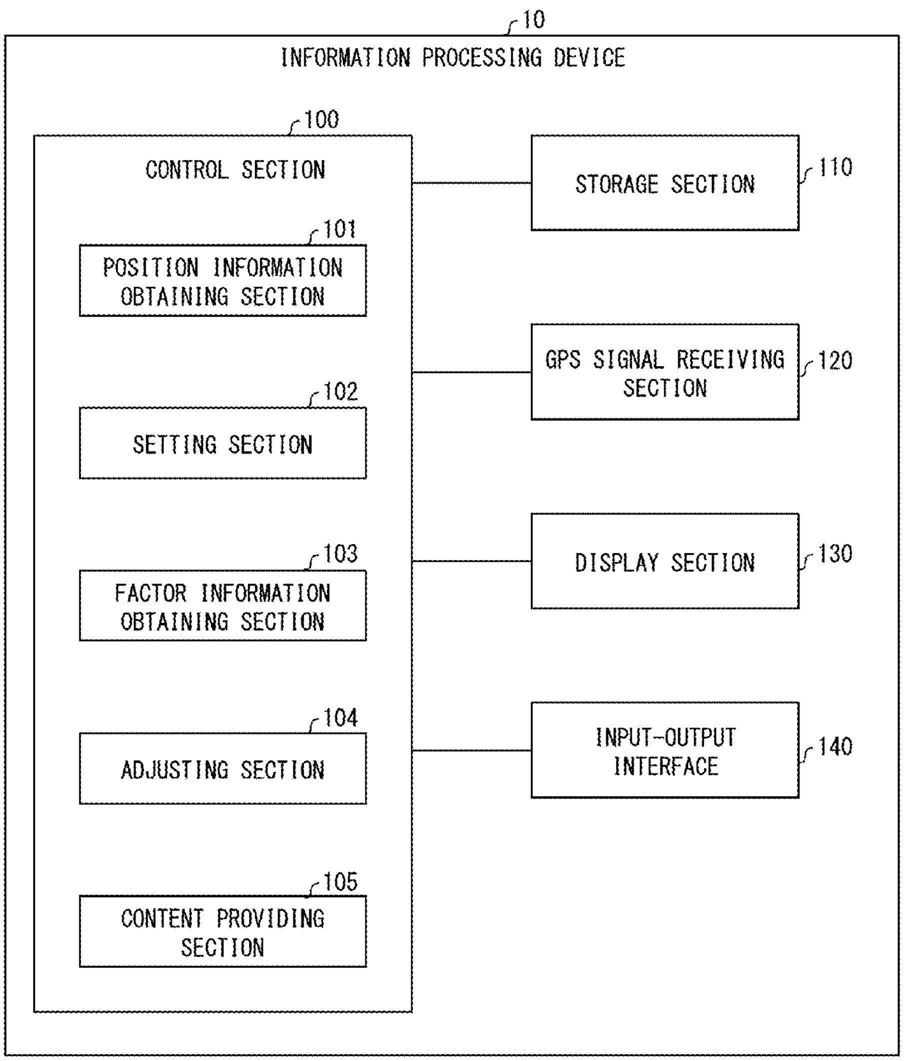


FIG. 1

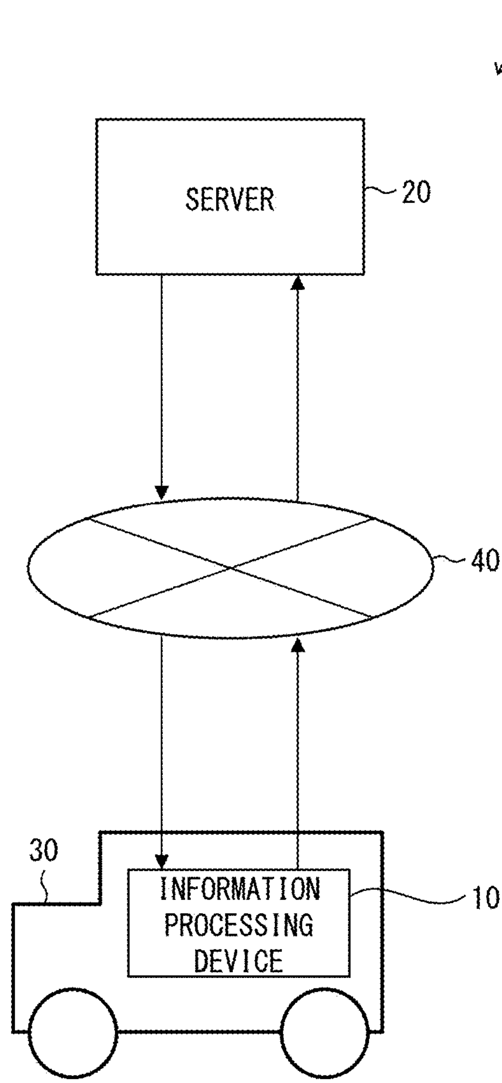


FIG. 2

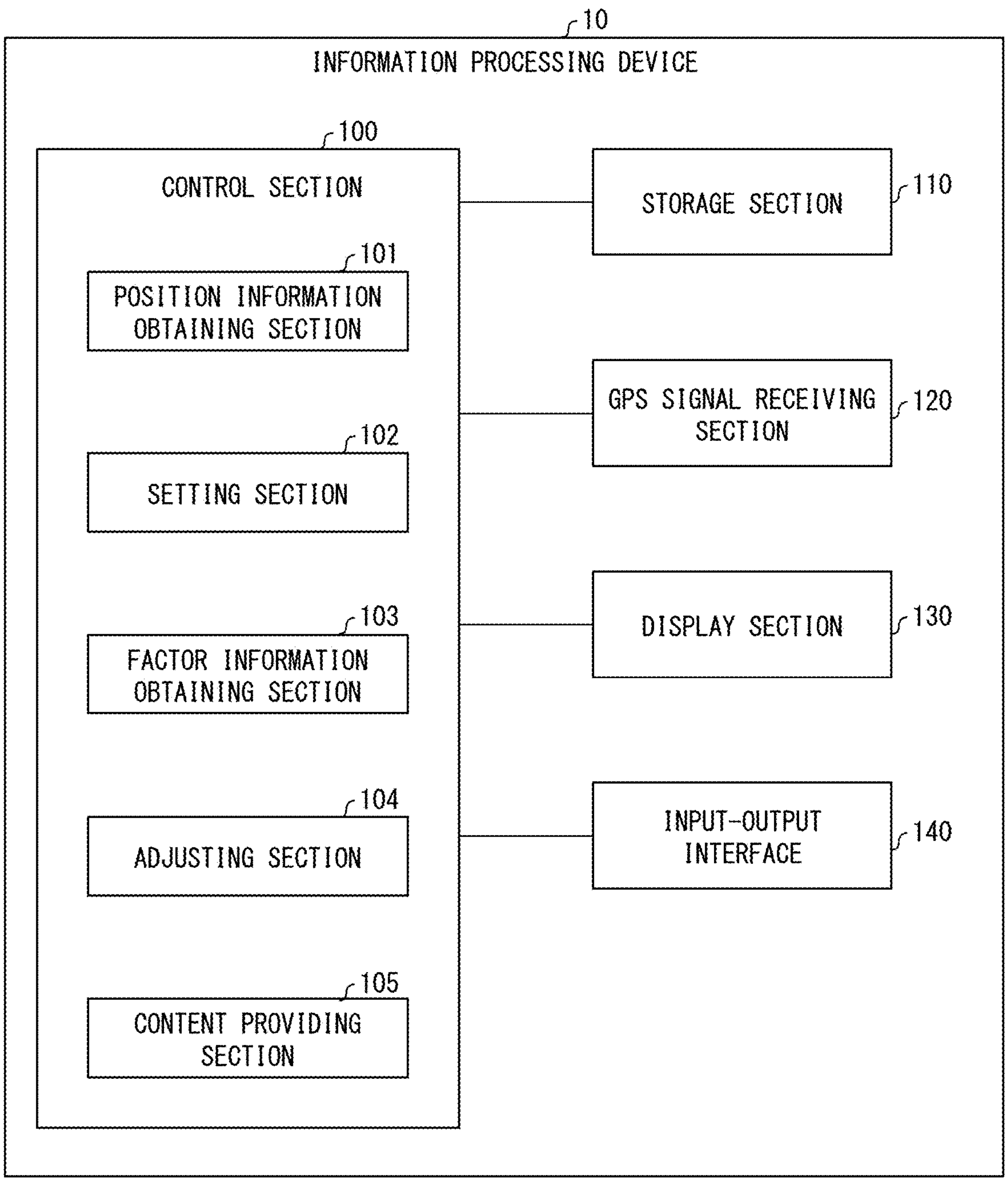

10
INFORMATION PROCESSING DEVICE
100
CONTROL SECTION
101
POSITION INFORMATION OBTAINING SECTION
102
SETTING SECTION
103
FACTOR INFORMATION OBTAINING SECTION
104
ADJUSTING SECTION
105
CONTENT PROVIDING SECTION
STORAGE SECTION
110
GPS SIGNAL RECEIVING SECTION
120
DISPLAY SECTION
130
INPUT-OUTPUT INTERFACE
140

FIG. 3

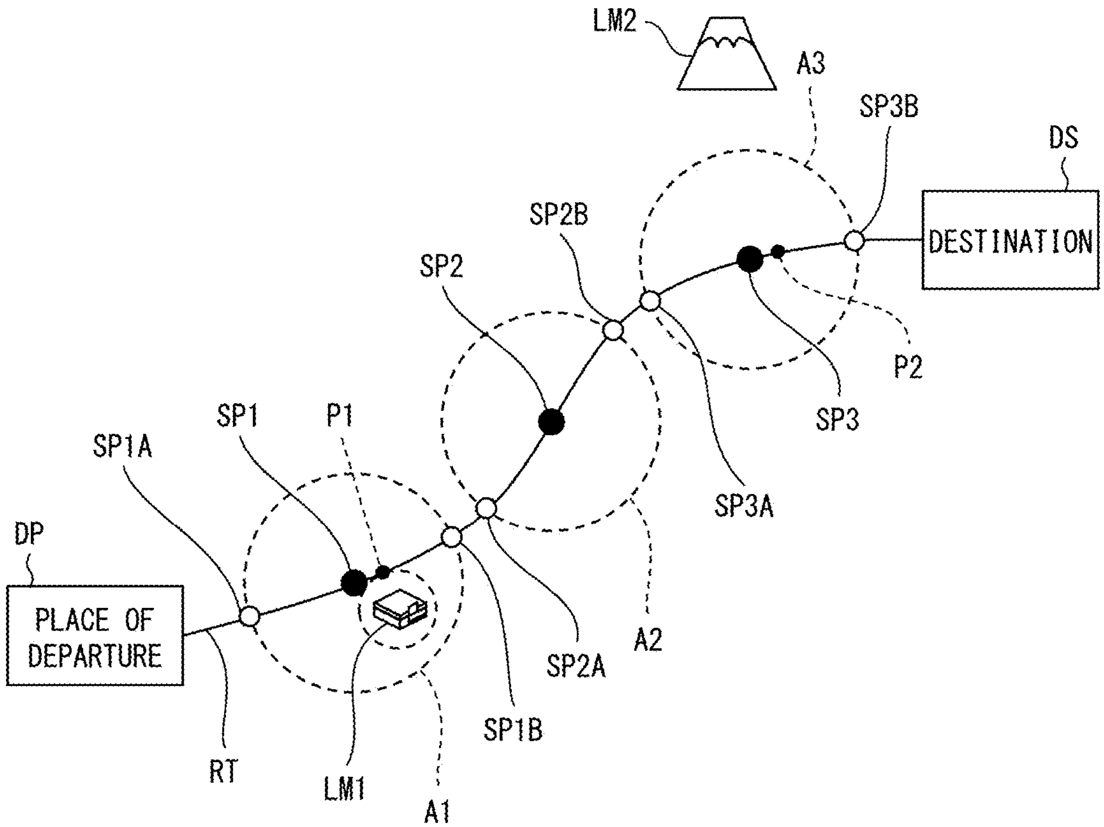
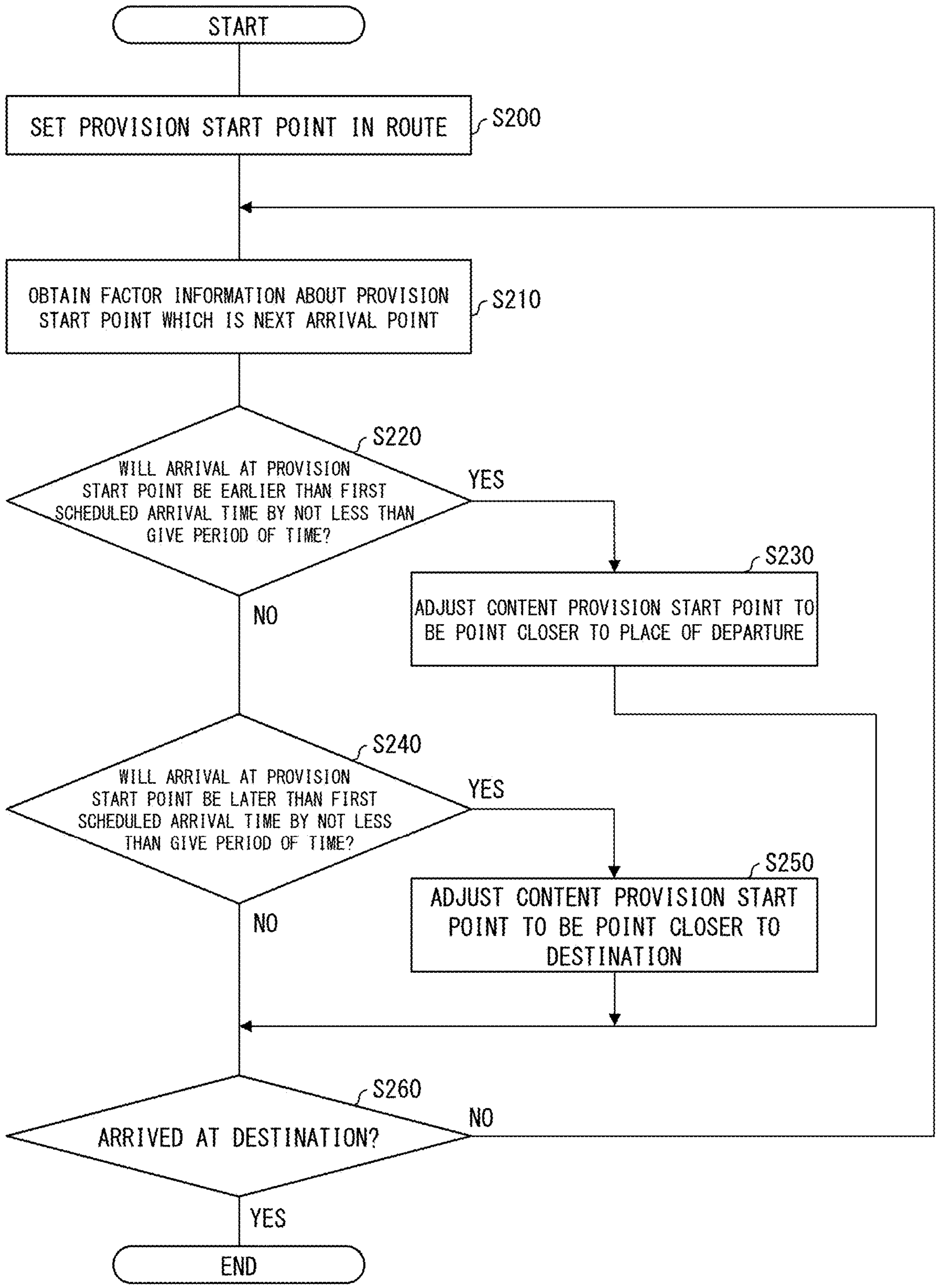

LM2
A3
SP3B
DS
DESTINATION
SP2B
SP2
P2
SP3
SP1
P1
SP1A
SP3A
DP
PLACE OF DEPARTURE
A2
SP2A
SP1B
RT
LM1
A1

FIG. 4


START
SET PROVISION START POINT IN ROUTE
S200
OBTAIN FACTOR INFORMATION ABOUT PROVISION START POINT WHICH IS NEXT ARRIVAL POINT
S210
WILL ARRIVAL AT PROVISION START POINT BE EARLIER THAN FIRST SCHEDULED ARRIVAL TIME BY NOT LESS THAN GIVE PERIOD OF TIME?
S220
YES
NO
ADJUST CONTENT PROVISION START POINT TO BE POINT CLOSER TO PLACE OF DEPARTURE
S230
WILL ARRIVAL AT PROVISION START POINT BE LATER THAN FIRST SCHEDULED ARRIVAL TIME BY NOT LESS THAN GIVE PERIOD OF TIME?
S240
YES
NO
ADJUST CONTENT PROVISION START POINT TO BE POINT CLOSER TO DESTINATION
S250
ARRIVED AT DESTINATION?
S260
NO
YES
END

INFORMATION PROCESSING DEVICE, CONTENT PROVIDING METHOD, VEHICLE, AND PROGRAM

[0001] This Nonprovisional application claims priority under 35 U.S.C. § 119 on Patent Application No. 2024-036167 filed in Japan on Mar. 8, 2024, the entire contents of which are hereby incorporated by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an information processing device, a content providing method, a vehicle, and a program.

BACKGROUND ART

[0003] Patent Literature 1 discloses a content distribution system for providing, to a passenger in a traveling body, content according to a destination when the passenger travels to the destination.

CITATION LIST

Patent Literature

[Patent Literature 1]

[0004] International Publication No. WO 2023/058501

SUMMARY OF INVENTION

Technical Problem

[0005] For providing content while a traveling body is traveling, a timing to provide the content is sometimes important. For example, even when content is provided to a passenger after a traveling body has passed through a place relating to the content, an effect aimed by the content may not be attained.

[0006] Time when a traveling body arrives at a given position may sometimes become earlier or later than scheduled time depending on various factors. For example, in cases of vehicles such as automobiles, buses, and taxies, it is considered that the time may become earlier or later than scheduled time due to a factor(s) such as a road condition and/or weather. In cases of buses, railroad trains, aircrafts, ships, and the like, it is considered that the time may become earlier or later than schedule time due to a factor(s) such as occurrence of an accident causing injury or death.

[0007] An aspect of the present disclosure has an object to provide content to a passenger of a traveling body at a targeted timing.

Solution to Problem

[0008] In order to attain the above object, an information processing device in accordance with an aspect of the present disclosure is an information processing device that provides content to a passenger of a traveling body traveling along a route, the information processing device including: a setting section configured to set, in the route, a provision start point where provision of the content is started; a factor information obtaining section configured to obtain, while the traveling body is traveling along the route, factor information about a factor that gives an influence on time when the traveling body arrives at the provision start point; and an adjusting section configured to adjust, while the traveling body is traveling along the route, the provision start point on a basis of the factor information.

[0009] In order to attain the above object, a content providing method in accordance with another aspect of the present disclosure is a content providing method for controlling an output device to provide content to a passenger of a traveling body traveling along a route, the content providing method including the steps of: setting, in the route, a provision start point where provision of the content is started; obtaining, while the traveling body is traveling along the route, factor information about a factor that gives an influence on time when the traveling body arrives at the provision start point; and adjusting, while the traveling body is traveling along the route, the provision start point on a basis of the factor information.

[0010] An information processing device in accordance with each aspect of the present disclosure may be realized by a computer. In this case, the present disclosure also encompasses: a program for the information processing device, the program causing a computer to operate as the sections (software elements) included in the information processing device so that the information processing device is realized by the computer; and a computer-readable storage medium in which the program is stored.

Advantageous Effects of Invention

[0011] According to an aspect of the present disclosure, it is possible to provide content to a passenger of a traveling body at a targeted timing.

BRIEF DESCRIPTION OF DRAWINGS

[0012] FIG. **1** is a block diagram illustrating an example of a configuration of a content providing system that uses an information processing device in accordance with an embodiment of the present disclosure.

[0013] FIG. **2** is a view illustrating an example of a configuration of the information processing device in accordance with the embodiment of the present disclosure.

[0014] FIG. **3** is a view used to explain a setting section and an adjusting section.

[0015] FIG. **4** is a flowchart relating to a content providing method in accordance with an embodiment of the present disclosure.

DESCRIPTION OF EMBODIMENTS

[0016] FIG. **1** is a block diagram illustrating an example of a configuration of a content providing system that uses an information processing device in accordance with an embodiment of the present disclosure. A content providing system **1** shown in FIG. **1** includes an information processing device **10** and a server **20**. For example, the information processing device **10** is disposed inside a vehicle **30**, and provides content to a passenger of the vehicle **30**. The vehicle **30** is one example of the traveling body, and is, for example, an automobile, a bus, or a taxi. The content is the one that can be sensed by the passenger with five senses. The content includes a video, an audio, a vibration, an odor, and/or mist, for example.

[0017] The server **20** is disposed outside the information processing device **10**. The server **20** has a function to transmit, to the information processing device **10**, data of the content that the information processing device **10** provides to the passenger. The data of the content includes: guiding content for providing information about (i) a facility existing in an area surrounding a road on which the vehicle **30** travels, (ii) a building that is viewable through a window of the vehicle **30**, and/or (iii) a landmark(s) such as a river and/or a mountain; and advertising content for advertisement.

[0018] The data relating to the guiding content includes (i) information indicative of a position (a latitude, a longitude, and/or an altitude) of a subject to be guided (i.e., a guiding subject) such as a facility and/or a landmark, (ii) data indicative of a video and/or an audio relating to the guiding subject, and (iii) information about a playback time of a video and/or an audio.

[0019] The information processing device **10** and the server **20** are connected to a network **40** such as a wireless communication network, an Internet network, or a telephone line network. For convenience of explanation, FIG. **1** illustrates a single information processing device **10**. Alternatively, the content providing system **1** may include a plurality of information processing devices **10**. Further, the information processing device **10** and the server **20** may be connected to, via the network **40** and/or the like, an information terminal, a server, and/or the like (not illustrated). For example, the information processing device **10** and the server **20** may be connected to an information terminal possessed by the passenger of the vehicle **30**.

[0020] FIG. **2** is a view illustrating an example of a configuration of the information processing device in accordance with the embodiment of the present disclosure. As shown in FIG. **2**, the information processing device **10** includes a control section **100**, a storage section **110**, a GPS signal receiving section **120**, a display section **130**, and an input-output interface **140**.

[0021] The control section **100** includes, e.g., a central processing unit (CPU) and a main storage device. The storage section **110** includes, e.g., a hard disk drive (HDD) and/or a solid state drive (SSD). In the storage section **110**, a program to be executed by the control section **100** and data of content transmitted from the server **20** are stored.

[0022] The GPS signal receiving section **120** receives a signal from a GPS satellite. The display section **130** is one example of the output device, and can be used to provide content such as a still image and/or a moving image. The input-output interface **140** is, for example, a universal serial bus (USB) terminal and/or a local area network (LAN) terminal. The information processing device **10** is connected, via the input-output interface **140**, to a vehicle communication network such as a controller area network (CAN). The information processing device **10** is connectable to the network **40** via CAN.

[0023] The information processing device **10** can obtain, via the input-output interface **140**, route information indicative of a route along which the vehicle **30** is scheduled to travel. For example, the information processing device **10** may obtain, via the network **40**, route information that the passenger has registered in the server **20** in advance. Further, the information processing device **10** may be configured such that (i) map data is stored in the storage section **110** and (ii) input of information about a place of departure and a destination from a user, such as a passenger, is accepted by a desired method and searching for a route from the place of departure to the destination is carried out.

[0024] The information processing device **10** obtains, from the server **20**, data of guiding content and advertisement content to be provided while the vehicle **30** is traveling along the route. In a case where the information processing device **10** obtains the route information from the server **20**, the information processing device **10** may obtain, from the server **20**, data of the content together with the route information. Further, in a case where the information processing device **10** searches a route, the information processing device **10** may transmit the route information to the server **20** via the network **40** and obtain the data of the content from the server **20**.

[0025] By executing the program stored in the storage section **110**, the control section **100** functions as a position information obtaining section **101**, a setting section **102**, a factor information obtaining section **103**, an adjusting section **104**, and a content providing section **105**.

[0026] The position information obtaining section **101** obtains position information indicative of a current position of the vehicle **30**. For example, the position information obtaining section **101** obtains, on the basis of a signal that the GPS signal receiving section **120** has received from a GPS satellite, position information indicative of a current position of the vehicle **30**. The position information obtaining section **101** may be connected to CAN via the input-output interface **140** and may be configured to obtain vehicle information about a vehicle speed of the vehicle **30**, an acceleration of the vehicle **30**, and amounts of operation of an accelerator pedal, a brake pedal, and a steering wheel of the vehicle **30** and to use the vehicle information to calculate the current position of the vehicle **30**.

[0027] FIG. **3** is a view used to explain the setting section and the adjusting section. FIG. **3** shows, on a route RT from a place of departure DP to a destination DS, positions of provision start points SP**1**, SP**2**, and SP**3** where provision of content is started. The route RT is a route along which the vehicle **30** is scheduled to travel. Each of the provision start points SP**1**, SP**2**, and SP**3** is one example of the provision start point.

[0028] In FIG. **3**, "LM**1**" indicates (i) a facility that is a guiding subject of the guiding content or (ii) a facility that deals with an advertisement subject of the advertisement content. The facility LM**1** is located at an area surrounding the route RT, and a position P**1** on the route RT is included within a range of a given distance from the facility LM**1**. Therefore, data such as guiding content relating to the facility LM**1** and advertisement content of an advertisement subject dealt with by the facility LM**1** is transmitted from the server **20** to the information processing device **10**. The "given distance" herein refers to, for example, a width of a road which is included in the route RT and which is located in an area surrounding the facility LM**1**.

[0029] A provision start point SP**1** indicates a position where provision of, e.g., the guiding content relating to the facility LM**1** is started. The setting section **102** sets scheduled time of arrival of the vehicle **30** at the provision start point SP**1** on the basis of, for example, information indicative of a position of the facility LM**1** and information about a playback time of the content. The scheduled time of arrival at the provision start point, which is set by the setting section **102**, will be referred to as "first scheduled time".

[0030] In FIG. **3**, "LM**2**" indicates a landmark which is the guiding subject of the guiding content. In an area surrounding a position P**2** on the route RT, a landmark LM**2** is sometimes viewable through a window of the vehicle **30**. Thus, data of guiding content relating to the landmark LM**2**

is transmitted from the server **20** to the information processing device **10**. The provision start point SP**3** indicates a position where provision of the guiding content relating to the landmark LM**2** is started. The setting section **102** sets first scheduled time when the vehicle **30** arrives at the provision start point SP**3**, for example, on the basis of information indicative of a position of the landmark LM**2** and information about a playback time of the guiding content.

[0031] The information processing device **10** starts providing each content at the first scheduled time of the content so as to make the passenger interested in the guiding subject and/or the advertisement subject. However, when the vehicle **30** actually moves along the route RT, time when the vehicle **30** arrives at the provision start point may sometimes become earlier or later than the first scheduled time due to various factors such as a road condition and/or weather.

[0032] The adjusting section **104** adjusts a position of the provision start point in consideration of various factors such as a road condition and/or weather that give an influence on the time when the vehicle **30** arrives as the provision start point. A provision area A**1** shown in FIG. **3** is a given-distance range centered on the provision start point SP**1**. The adjusting section **104** adjusts a position where provision of the guiding content relating to the facility LM**1** is started so that the position is set within the provision area A**1**. The "given distance" is 20 m, for example. The "given distance" may vary depending on the provision start point SP**1**. For example, the given distance may vary depending on a road width, the number of lanes, and/or the like of the provision start point SP**1**.

[0033] In FIG. **3**, an outer circumference of the provision area A**1** intersects the route RT at two points. A provision start point SP**1**A is provided at an intersecting point closer to the place of departure DP than the provision start point SP**1** is, and a provision start point SP**1**B is provided at an intersecting point closer to the destination DS than the provision start point SP**1** is. For example, in a case where there is a possibility that the vehicle **30** may arrive at the provision start point SP**1** earlier than the first scheduled time, the adjusting section **104** adjusts the provision start point SP**1** so that the provision start point SP**1** is set at the provision start point SP**1**A. For another example, in a case where there is a possibility that the vehicle **30** may arrive at the provision start point SP**1** later than the first scheduled time, the adjusting section **104** adjusts the provision start point SP**1** so that the provision start point SP**1** is set at the provision start point SP**1**B.

[0034] The adjusting section **104** adjusts a position where provision of the guiding content relating to the landmark LM**2** is started so that the position is set within a provision area A**3** centered on the provision start point SP**3**. In FIG. **3**, an outer circumference of the provision area A**3** intersects the route RT at two points. A provision start point SP**3**A is provided at an intersecting point closer to the place of departure DP than the provision start point SP**3** is, and a provision start point SP**3**B is provided at an intersecting point closer to the destination DS than the provision start point SP**3** is. For example, in a case where there is a possibility that the vehicle **30** may arrive at the provision start point SP**3** earlier than the first scheduled time, the adjusting section **104** adjusts the provision start point SP**3** so that the provision start point SP**3** is set at the provision start point SP**3**A. For another example, in a case where there is a possibility that the vehicle **30** may arrive at the provision start point SP**3** later than the first scheduled time, the adjusting section **104** adjusts the provision start point SP**3** so that the provision start point SP**3** is set at the provision start point SP**3**B.

[0035] A provision start point SP**2** indicates a position where provision of the advertisement content is started. In a case where the route RT includes a partial path for which provision of the content is not scheduled, the setting section **102** sets, on the partial path, the provision start point for the advertisement content. FIG. **3** shows a provision area A**2** centered on the provision start point SP**2**. An outer circumference of the provision area A**2** intersects the route RT at two points. A provision start point SP**2**A is provided at an intersecting point closer to the place of departure DP than the provision start point SP**2** is, and a provision start point SP**2**B is provided at an intersecting point closer to the destination DS than the provision start point SP**2** is. For example, in a case where there is a possibility that the vehicle **30** may arrive at the provision start point SP**2** earlier than the first scheduled time, the adjusting section **104** adjusts the provision start point SP**2** so that the provision start point SP**2** is set at the provision start point SP**2**A. For another example, in a case where there is a possibility that the vehicle **30** may arrive at the provision start point SP**2** later than the first scheduled time, the adjusting section **104** adjusts the provision start point SP**2** so that the provision start point SP**2** is set at the provision start point SP**2**B.

[0036] While the vehicle **30** is traveling along the route RT, the factor information obtaining section **103** shown in FIG. **2** obtains factor information about (i) a factor that gives an influence on time when the vehicle **30** arrives at the provision start point SP**1** or the like and (ii) a factor that causes the time when the vehicle **30** arrives at the provision start point to become earlier or later than the first scheduled time.

[0037] For example, with regard to the position of the vehicle **30** obtained by the position information obtaining section **101**, the factor information obtaining section **103** may obtain, as the factor information, a difference between scheduled arrival time and actual arrival time. For example, in a case where the actual arrival time is earlier than the scheduled arrival time by not less than a given period of time, the difference between the scheduled arrival time and the actual arrival time serves as factor information indicative of a factor that causes time when the vehicle **30** arrives at the provision start point to become earlier than the first scheduled time. In a case where the actual arrival time is later than the scheduled arrival time for not less than a given period of time, the difference between the scheduled arrival time and the actual arrival time serves as factor information indicative of a factor that causes the time when the vehicle **30** arrives at the provision start point becomes later than the first scheduled time.

[0038] Further, the factor information obtaining section **103** may access, via network **40**, a web service that provides information about a road condition and/or weather and may obtain, as the factor information, information about a road condition and/or weather of an area surrounding the provision start point.

[0039] When the current position of the vehicle **30** obtained by the position information obtaining section **101** reaches the provision start point set by the adjusting section **104**, the content providing section **105** provides content corresponding to the provision start point. For example, in a case where the content is a video, the content providing section **105** plays back the video via the display section **130** of the information processing device **10**. The content providing section **105** may play back the video via the display section provided in the vehicle **30** and/or an information terminal possessed by the passenger on the vehicle **30**.

[0040] FIG. **4** is a flowchart relating to a content providing method in accordance with an embodiment of the present disclosure. The process shown in FIG. **4** is executed by the control section **100** while the vehicle **30** is traveling along the route. In the description below, it is assumed that the vehicle **30** is traveling along the route RT shown in FIG. **3**.

[0041] In S**200**, the control section **100** functions as the setting section **102** to set, in a route along which the vehicle **30** is scheduled to travel, a provision start point where provision of content is started. For example, as shown in FIG. **3**, the control section **100** sets, in the route RT along which the vehicle **30** is scheduled to travel, the provision start points SP**1**, SP**2**, and SP**3** where provision of content is started.

[0042] In subsequent S**210**, the control section **100** obtains factor information about a provision start point at which the vehicle **30** will arrive at next. For example, in a case where the vehicle **30** is traveling in an area near the place of departure DP, the control section **100** obtains factor information about the provision start point SP**1**, at which the vehicle **30** will arrive at next.

[0043] In S**220**, the control section **100** determines, on the basis of the factor information obtained in S**210**, whether or not the vehicle **30** will arrive at the provision start point SP**1** at an earlier timing. For example, in a case where, with regard to the position of the vehicle **30** obtained by the position information obtaining section **101**, actual arrival time is earlier than scheduled arrival time by not less than a given period of time, the control section **100** determines that the vehicle **30** will arrive at the provision start point SP**1** at an earlier timing (S**220**: YES), and adjusts the provision start point SP**1** for the content so that the provision start point SP**1** is set at the provision start point SP**1**A, which is located closer to the place of departure DP (S**230**). Meanwhile, in a case where, with regard to the position of the vehicle **30** obtained by the position information obtaining section **101**, the actual arrival time is not earlier than the scheduled arrival time by not less than a given period of time (S**220**: NO), the control section **100** advances to a process in S**240**.

[0044] In S**240**, on the basis of the factor information obtained in S**210**, the control section **100** determines whether or not the vehicle **30** will arrive at the provision start point SP**1** at a later timing. For example, in a case where, with regard to the position of the vehicle **30** obtained by the position information obtaining section **101**, the actual arrival time is later than the scheduled arrival time by not less than the given period of time, the control section **100** determines that the vehicle **30** will arrive at the provision start point SP**1** at a later timing (S**240**: YES), and adjusts the provision start point SP**1** for the content so that the provision start point SP**1** is set at the provision start point SP**1**B, which is located closer to the destination DS (S**250**). In a case where, with regard to the position of the vehicle **30** obtained by the position information obtaining section **101**, the actual arrival time is not later than the scheduled arrival time by not less than the given period of time (S**240**: NO), the control section **100** advances to a process in S**260**.

[0045] In S**260**, the control section **100** determines whether or not the vehicle **30** has arrived at the destination DS. In a case where the vehicle **30** has arrived at the destination DS (S**260**: YES), the control section **100** ends the processes shown in FIG. **4**. Meanwhile, in a case where the vehicle **30** has not arrived at the destination DS (S**260**: NO), the control section **100** advances to a process in S**210**.

Effects

[0046] As explained above, according to the information processing device **10** in accordance with the present embodiment, it is possible to attain the following effects.

[0047] The information processing device **10** provides content to the passenger of the vehicle **30** traveling along the route RT, and includes the setting section **102**, the factor information obtaining section **103**, and the adjusting section **104**. The setting section **102** sets, in the route RT, the provision start point SP**1** or the like where provision of the content is started. The factor information obtaining section **103** obtains, while the vehicle **30** is traveling along the route RT, factor information about a factor that gives an influence on time when the vehicle **30** arrives at the provision start point SP**1** or the like. While the vehicle **30** is traveling along the route RT, the adjusting section **104** adjusts the provision start point SP**1** on the basis of the factor information. With the above configuration, the information processing device **10** can provide the content to the passenger of the vehicle **30** at a targeted timing.

[0048] In a case where the factor information obtaining section **103** obtains information about a factor that causes the time when the vehicle **30** arrives at the provision start point SP**1** or the like to become earlier, the adjusting section **104** adjusts the provision start point SP**1** or the like so that the provision start point SP**1** or the like is set at a position (e.g., the provision start point SP**1**A) closer to the place of departure DP in the route RT. In a case where the factor information obtaining section **103** obtains information about a factor that causes the time when the vehicle **30** arrives at the provision start point SP**1** or the like to become later, the adjusting section **104** adjusts the provision start point SP**1** or the like so that the provision start point SP**1** or the like is set at a position (e.g., the provision start point SP**1**B) closer to the destination DS in the route RT. With the above configuration, the information processing device **10** can provide the content at a timing at which the vehicle **30** can arrive at the provision start point.

[0049] The information processing device **10** further includes the position information obtaining section **101** configured to obtain position information indicative of a current position of the vehicle **30**. The factor information obtaining section **103** obtain factor information on the basis of a difference between (i) time when the vehicle **30** is scheduled to arrive at the current position and (ii) time when the vehicle **30** actually arrived at the current position, each obtained in a case where the vehicle **30** travels along the route RT. With the above configuration, the vehicle **30** can obtain, through the position information of the vehicle **30** and/or the like, an accurate factor(s) associated with various matters such as a road condition and/or weather.

[0050] The content includes content for providing, to the passenger, information about a given guiding subject. The guiding subject includes (i) a subject (e.g., the facility LM**1**) located within a range of a given distance from the route RT and (ii) a subject (e.g., the landmark LM**2**) viewable from the route RT. With the above configuration, by providing the content regarding the guiding subject, it is possible to uplift the passenger's mood while the vehicle **30** is traveling along the route RT.

[0051] The content providing method shown in FIG. **4** controls the display section **130** to provide the content to the passenger of the vehicle **30** traveling along the route RT. The content providing method includes the steps of: setting, in the route RT, the provision start point SP**1** or the like where provision of the content is started (S**200**); obtaining, while the vehicle **30** is traveling along the route RT, the factor information about a factor that gives an influence on time when the vehicle **30** arrives at the provision start point SP**1** or the like (S**210**); and adjusting, while the vehicle **30** is traveling along the route RT, the provision start point SP**1** or the like on the basis of the factor information (S**220** to S**250**). With the above configuration, the content providing method shown in FIG. **4** can provide the content to the passenger of the vehicle **30** at a targeted timing.

[0052] The vehicle **30** includes the information processing device **10** and the output device configured to output the content.

Software Implementation Example

[0053] The functions of the information processing device **10** (hereinafter, referred to as a "device") can be realized by a program for causing a computer to function as the device, the program causing the computer to function as the control blocks (particularly, the sections included in the control section **100**) of the device.

[0054] In this case, the device includes, as hardware for executing the program, at least one control device (e.g., a processor) and at least one storage device (e.g., a memory). By causing the control device and the storage device to execute the program, the functions explained in the foregoing embodiments are realized.

[0055] The program may be recorded in one or more non-transitory computer-readable recording media. The recording media may or may not be included in the device. In the latter case, the program may be supplied to or made available to the device via any wired or wireless transmission medium.

[0056] Furthermore, some or all of the functions of the control blocks can also be realized by a logic circuit. For example, the present disclosure encompasses, in its scope, an integrated circuit in which a logic circuit that functions as each of the above-described control blocks is formed. In addition, the function of each of the control blocks can be realized by, for example, a quantum computer.

[0057] The processes described in the foregoing embodiments may be carried out by artificial intelligence (AI). In this case, AI may be operated in the control device, or may be operated in another device (e.g., an edge computer or a cloud server).

Variations

[0058] The foregoing embodiment has dealt with the content providing method that provides content while the vehicle **30** is traveling on a road. However, the information processing device and the content providing method in accordance with one embodiment of the present disclosure are applicable to provision of content carried out during traveling of railroad trains, aircrafts, ships, and the like that do not travel on a road.

[0059] In the foregoing embodiment, the information processing device **10** is configured such that data of content that is to be provided to a passenger is transmitted from f the server **20** to the information processing device **10**. Alternatively, however, the data of the content may be stored in the storage section **110** of the information processing device **10** in advance.

[0060] In the foregoing embodiment, the control section **100** of the information processing device **10** provided inside the vehicle **30** functions as the setting section **102**, the factor information obtaining section **103**, and the adjusting section **104**. Alternatively, however, the information processing device including the setting section **102**, the factor information obtaining section **103**, and the adjusting section **104** may be provided outside the vehicle **30**. For example, the server **20** may function as the information processing device. In a case where the server **20** is caused to function as the setting section **102**, the factor information obtaining section **103**, and the adjusting section **104**, (i) position information of the vehicle **30** and/or (ii) vehicle information about a vehicle speed, an acceleration, and/or an amount of operation of an accelerator pedal of the vehicle **30** may be transmitted from the information processing device **10** to the server **20**.

SUPPLEMENTARY REMARKS

[0061] The present disclosure is not limited to the embodiments above, but can be altered by a skilled person in the art within the scope of the claims. The present disclosure also encompasses, in its technical scope, any embodiment derived by combining technical means disclosed in differing embodiments as appropriate.

REFERENCE SIGNS LIST

[0062] **10**: information processing device
[0063] **20**: server
[0064] **30**: vehicle
[0065] **40**: network
[0066] **100**: control section
[0067] **101**: position information obtaining section
[0068] **102**: setting section
[0069] **103**: factor information obtaining section
[0070] **104**: adjusting section
[0071] SP**1**, SP**1**A, SP**1**B, SP**2**, SP**2**A, SP**2**B, SP**3**, SP**3**A, SP**3**B: provision start point (provision start point)

1. An information processing device that provides content to a passenger of a traveling body traveling along a route, the information processing device comprising:

a setting section configured to set, in the route, a provision start point where provision of the content is started;

a factor information obtaining section configured to obtain, while the traveling body is traveling along the route, factor information about a factor that gives an influence on time when the traveling body arrives at the provision start point; and

an adjusting section configured to adjust, while the traveling body is traveling along the route, the provision start point on a basis of the factor information.

2. The information processing device according to claim **1**, wherein:

in a case where the factor information obtaining section obtains information about a factor that causes the time when the traveling body arrives at the provision start point to become earlier, the adjusting section adjusts the provision start point so that the provision start point is set at a position closer to a place of departure in the route; and

in a case where the factor information obtaining section obtains information about a factor that causes the time when the traveling body arrives at the provision start point to become later, the adjusting section adjusts the provision start point so that the provision start point is set at a position closer to a destination in the route.

3. The information processing device according to claim **1**, further comprising:

a position information obtaining section configured to obtain position information indicative of a current position of the traveling body, wherein:

the factor information obtaining section obtains the factor information on a basis of a difference between (i) time when the traveling body is scheduled to arrive at the current position and (ii) time when the traveling body actually arrived at the current position, each obtained in a case where the traveling body travels along the route.

4. The information processing device according to claim **1**, wherein:

the content includes content for providing, to the passenger, information about a given guiding subject; and

the given guiding subject includes (i) a subject located within a range of a given distance from the route and (ii) a subject viewable from the route.

5. A content providing method for controlling an output device to provide content to a passenger of a traveling body traveling along a route, the content providing method comprising the steps of:

setting, in the route, a provision start point where provision of the content is started;

obtaining, while the traveling body is traveling along the route, factor information about a factor that gives an influence on time when the traveling body arrives at the provision start point; and

adjusting, while the traveling body is traveling along the route, the provision start point on a basis of the factor information.

6. A vehicle comprising:

an information processing device recited in claim **1**; and

an output device configured to output content.

7. A program for causing a computer to function as an information processing device recited in claim **1**, the program causing the computer to function as the setting section, the factor information obtaining section, and the adjusting section.

* * * * *